STUDENT PROJECT BRIEF

# Sunday Project | Data Quality, Insights & CSV Operations

Beginner-level Pandas project in Google Colab or Jupyter Notebook. Complete both parts in one runnable notebook.

**SUBMISSION RULES — MANDATORY**

ZIP FILE ONLY: Upload one .zip file containing all required assignment files. Individual files will not be accepted.
RENAME YOUR ZIP: Use StudentName_TrainingID.zip (replace both parts with your own name and training ID).
NO LATE SUBMISSIONS: Submit before the portal deadline. Late submissions will not be accepted.

## Part A | Data Quality & Basic Insights — 70 marks

Goal: Turn a raw sales CSV into cleaned data, one grouped summary, and two clear charts. Work with sales_raw.csv.

### 1. Load and inspect

- Read the CSV using pd.read_csv(). Show head(), shape, column names and data types.
- Check missing values with isna().sum() and duplicate rows with duplicated().sum().

### 2. Clean the dataset

- Trim extra spaces and standardize inconsistent text in region and category.
- Handle missing values, invalid prices and duplicates using sensible rules; explain the choices.
- Compare the row count before and after cleaning; preserve the original input CSV.

### 3. Calculate and analyze

- Add a revenue column: quantity × unit_price.
- Use groupby() to summarize revenue by region or category.
- Create two labelled charts and write two evidence-based findings.

### 4. Save the result

- Export a new cleaned_sales.csv using to_csv(..., index=False).

Part A output: Runnable notebook, cleaned_sales.csv, two charts and two written findings.

## Part B | CSV File Handling & Manipulation — 30 marks

Goal: Edit the CSV created in Part A and save a separate output called sales_updated.csv.

### 1. Read and add a column

• Load cleaned_sales.csv with pd.read_csv().

• Add tax_amount equal to 5% of revenue and show the first five rows.

### 2. Update and remove

• Use .loc[] to increase the unit_price of Books items by 10%; recalculate revenue and tax_amount.

• Remove one selected row using an existing transaction_id.

• Create a temporary review_status column, then remove it with drop(columns=...).

### 3. Save and verify

• Save to sales_updated.csv, read it again, and show the first five rows and its final column names.

Part B output: Runnable CSV-editing code and sales_updated.csv.

## What to put inside your ZIP

• One .ipynb notebook with clearly marked Part A and Part B sections.

• The input sales_raw.csv and output files cleaned_sales.csv and sales_updated.csv.

• A short README.md explaining how to run the notebook and containing your GitHub repository link.

## Submission checklist

☐ Put all required files together in a ZIP archive.

☐ Name it StudentName_TrainingID.zip using your own details.

☐ Upload the ZIP file only before the portal deadline. Late submissions will not be accepted.

☐ Maintain meaningful GitHub commits and include the repository URL in your README.

Marks: Part A 70 + Part B 30 = 100 total. Use relative paths so the notebook works in both Jupyter and Colab. SQL and Streamlit are optional extensions.